

Informal Settlement Detection and Growth Tracking in Peri-Urban Areas of Kolkata Using Multi-Temporal Satellite Imagery and Deep Learning

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Abstract

Rapid, unplanned urbanisation has driven significant growth of informal settlements in Kolkata's peri-urban fringe, encroaching on agricultural land, floodplains, and ecologically sensitive zones. Globally, informal settlements house over a billion people, yet their systematic mapping remains challenging due to spectral heterogeneity, spatial irregularity, and severe class imbalance relative to formal urban areas. Existing approaches are limited by the high cost of commercial satellite imagery, the inability of pixel-based classifiers to capture morphological complexity, and a lack of frameworks validated for South Asian cities. Accurate delineation of these zones is essential for targeted urban governance, public health intervention, climate resilience planning, and progress toward UN Sustainable Development Goal 11 on Sustainable Cities and Communities.

This project proposes an end-to-end pipeline for detecting, tracking, and forecasting informal settlement growth in the Kolkata Metropolitan Area using freely available multi-temporal satellite imagery. The objectives are: (i) build a labelled geospatial dataset combining Sentinel-1 SAR, Sentinel-2 multi-spectral time series, and OpenStreetMap layers; (ii) develop a U-Net-based deep learning segmentation model for pixel-wise settlement delineation at 10 m resolution; (iii) perform multi-decadal temporal change detection (1997–present) to map directional growth and generate a five-year probabilistic expansion forecast; and (iv) deploy the trained model in an interactive GIS dashboard for municipal planners.

The methodology involves acquiring and preprocessing Sentinel-1/2 imagery via Google Earth Engine, deriving texture and spectral indices (GLCM, NDBI, NDVI), and training a U-Net model on annotated patches. Given that informal settlements occupy only 5–15% of urban land area, class imbalance is addressed using the Boundary-Aware Label-Imbalanced Settlement Estimation (BALISE) technique combined with SMOTE and Mixup-based data augmentation to improve model generalisation on minority-class samples. The trained model is applied retrospectively across the historical archive, with Markov chain transition modelling used to quantify directional growth patterns and produce the five-year expansion probability surface. Outputs are validated through spatial cross-validation and transferability testing on a secondary city.

Expected outcomes include a segmentation model achieving an F1 score ≥ 0.80 , settlement probability maps at 10 m resolution, a five-year growth forecast surface correlated with socioeconomic poverty indices, uncertainty maps for field verification, and a policy brief identifying priority intervention zones. This work aims to deliver a replicable, low-cost computational pipeline for informal settlement monitoring, contributing both to geospatial machine learning research and to urban governance in Kolkata.

Keywords: informal settlements, remote sensing, U-Net, Sentinel-2, temporal change detection, peri-urban growth, BALISE, SMOTE, Mixup, Markov chain modelling, GIS dashboard, Kolkata